

Data-Driven Prediction of Protein Unfolding Kinetics

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Abstract

This project aims to leverage existing biological knowledge of protein folding and unfolding dynamics by utilizing data from MechanoProDB, a web-based database for exploring the mechanical properties of proteins. The approach combines biophysical data obtained from single-molecule force spectroscopy experiments with advanced Machine Learning techniques, primarily consisting of transformer-based models. Among these are ESM2 with 8M parameters, and the combination of ProtBERT, pre-trained on 2.1B protein sequences, with SciBERT, which incorporates domain-specific language collected from 1.14M scientific papers. The research focuses on how experimental conditions, including pH, temperature, mechanical forces and denaturants, interact with different types of proteins to influence unfolding behavior and kinetic changes in proteins' stability. By learning these relationships, this project seeks to build the optimal model architecture capable of predicting mechanical properties of proteins under different scenarios. This is a novel method, enabling scientists to get approximate kinetic parameters and protein unfolding spectra, based on their specific conditions, not requiring laboratory experiments.

Keywords: artificial intelligence; machine learning; neural networks; protein kinetics; biophysics

Introduction

In recent years, rapid advancements in technology caused a significant progress in the field of Artificial Intelligence (AI), giving rise to many specialized subfields including Machine Learning

(ML), Deep Learning (DL), Natural Language Processing (NLP), Generative AI, among others. Emergence of these fields, in turn, enhanced plenty of other domains, determining the future of a number of unresolved challenges with modern technologies, complex algorithms and increased available computational resources. This opened up space for new research and findings, which could possibly broaden the scope of scientific discoveries and contribute to breakthroughs in every sphere.

Advancements in biology and physics due to this progress are crucial for understanding structural composition of living organisms and mechanisms behind their functional behavior at all levels. These developments allow scientists to investigate diverse processes including protein folding and unfolding with high precision and accuracy. Integration of AI into these fields allows experiments not possible before due to lack of resources, time complexity and microscopic scale of subjects.

Proteins are composed of long chains of amino acids, organic compounds that contain both amino and carboxylic acid functional groups, forming a vast number of conformations. However, not all of them are functional in their folded state. Predicting and tracking these conformational changes is difficult because folding and unfolding of proteins occurs on extremely short timescales (less than a second), involving intramolecular interactions governed by thermodynamics and quantum-level forces.

Conventional modern experimental methods for analyzing protein folding and unfolding states include X-ray crystallography, Fluorescence spec-

troscopy, Circular dichroism (CD), Protein nuclear magnetic resonance (NMR), Mass spectrometry, Fluorescence Resonance Energy Transfer (FRET), Atomic Force Microscopy (AFM), and some others have made it possible to obtain detailed information about the diverse conformational states occurring in the folding process of proteins. However, aforementioned limitations of classic ways of analysis mostly refer to all mentioned traditional techniques.

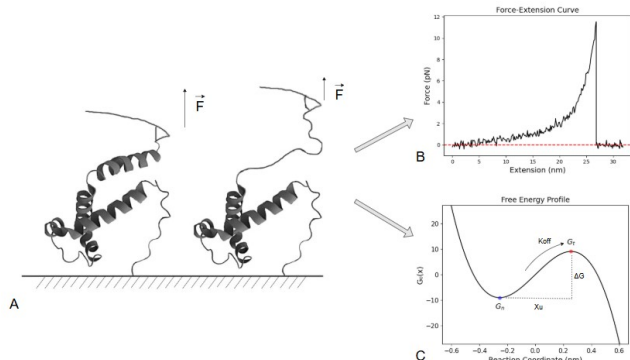


Figure 1: A. General image of protein unfolding through AFM (protein is fixed by cantilever, through which force is applied, leading to domains unfolding, depicted as a second state); B. Energy Landscape Profile obtained after model predictions during the inference; C. Force-Extension Curve obtained via predictions of the model during the inference stage.

Recent advancements in AI enabled scientists and researchers to predict the structure of proteins from their sequences (ex: AlphaFold [8], ESM-Fold [12]) along with existing protein data banks (ex: PDB [2], UniProt), being constantly updated, deprived them of the necessity to conduct high-cost experiments to obtain the protein details about their structure, 3D coordinates along with solvents and binding molecules, functional information and some theoretical annotation about family of organisms and similarity scores. These achievements are widely used in drug discovery, personalized medicine, genetic engineering and other fields by advancing computational modeling and simulations without being directly involved in real-life experiments and not requiring special equipment (except for computational resources if models are sufficiently large).

Thereby, this project aims to predict protein kinetics based on protein sequence and the

experimental conditions, such as concentration of solutes, extremes of pH, mechanical forces, the presence of chemical denaturants, and others, which potentially affect the mechanical stability of the proteins. Understanding protein stability and the unfolding pathway is crucial in the scientific field as proteins lose their biological activity if they are not sufficiently stable and may easily be disrupted by natural or external processes, including heat and occurring mutations in the sequences. In this project, ML models will be utilized and fine-tuned for solving this relevant problem of predicting the stability of proteins and their functional potential.

Literature Review

Mechanochemistry and SMFS Techniques

This project employs the idea of protein unfolding, making it essential to start the research from the core ideas behind the mechanochemistry and traditional experimental methods used for protein analysis, particularly AFM and Optical Tweezers.

Firstly, it should be noted that mechanochemistry itself is the study of chemical transformations induced by mechanical stress, which applies to protein unfolding. This field includes molecular biophysics, uncovering how the application of force alters reaction rates and molecular behavior. That means mechanical force directly modifies energy landscapes, thus shifting the thermodynamic equilibrium and kinetic behavior of molecular systems. As a base, single-molecule force spectroscopy (SMFS) techniques, which include Optical Tweezers and AFM, are highly utilized to observe force-dependent reaction kinetics. Applying extended models, capable of capturing molecular compliance, serving as more nuanced models, is more preferable than previously applied Eyring-Zhurkov-Bell (EZB) model, failing in case of molecular structures' deformation under high force [14]. As a result, the multidimensionality of molecular energy landscapes is highlighted, making it ineffective to use one-dimensional models due to employment of fixed transition states, while in reality transition states can shift unpredictably, even changing their direction to opposite towards applied force.

Following the theoretical foundation laid by Makarov [14], Noy [18] presented a more practical approach, analyzing the ground behind the AFM, providing a full guide to understanding and using this technique. AFM technology, with its piezoelectric scanners and cantilever-based force detection, plays a significant role in observing protein unfolding by experimenting with several approaches. The major finding is that rupture force depends on both the loading rate and the cantilever stiffness. With higher loading rate, the entire system quickly crosses the energy barrier, enabling scientists to measure the distance to the transition state. In contrast, lower loading rate allows fluctuations between bound and unbound states due to which equilibrium free energy can be measured under specific conditions. That means that the absolute value of the rupture force is meaningful only when the experimental conditions are known and clearly defined, as variations in those can lead to different results.

On the other hand, Bustamante in his paper [4] described another technique, widely used for measuring protein unfolding kinetics. Optical Tweezers allow researchers to observe and manipulate single molecular events in real time. They use focused laser beams to trap microscopic beads attached to molecules, which enable the measurement and application of different force range and displacement with high precision. Along with the usual utilization of Optical Tweezers, they are also combined with fluorescence, allowing simultaneous mechanical and visual analysis of molecular dynamics. That means that protein unfolding can be observed and measured to build the approximate energy landscape and understand the mechanisms behind biomolecular interactions between proteins.

Protein Unfolding

After that, preliminary research has been conducted in the field of protein unfolding to unravel the fundamental mechanisms behind how and why proteins unfold, what intermediate states they pass through, and the key factors that mainly influence this process. Understanding these dynamics is crucial, as it provides insights into protein stability, determining both the longevity of the designed proteins and their behavior under

various physical and chemical conditions. Losing structural integrity by unfolding can cause loss of proteins' functionality along with emergence of toxicity, which is highly undesirable by scientists.

Proteins can unfold under two main conditions: through the application of mechanical forces - such as those used in Atomic Force Microscopy (AFM) and Optical Tweezers - or via chemical denaturation. Fundamental differences have been observed between them, while analyzing unfolding pathways of proteins using single-molecule force spectroscopy and molecular dynamics simulations of ubiquitin. Mechanically unfolded proteins adopt fully extended conformations losing all secondary and tertiary structures and behaving like worm-like chains. In contrast, chemical denaturation with agents like urea typically produces only partially unfolded structures, retaining some elements of the original conformation [25].

Besides the approach used for protein unfolding, the direction of pulling significantly influences the unfolding trajectories, whether the protein was pulled from C-terminus, N-terminus or intermediary amino acids. Analyzing the molecular anisotropy of proteins, it was uncovered that when pulling is conducted from the C-terminus, the first unfolding event typically occurred in the module closest to the pulling point, matching the theoretical expectations of the observers. However, when pulling is done from N-terminus, unfolding takes unpredictable forms and patterns. Despite these differences, the geometric criteria used to classify unfolding remained valid for both pulling directions [22].

Alongside, not only the direction of pulling matters but also the velocity. The authors [23] have specified two types of velocities, v_c which is a critical pulling speed and v_p , being the pulling velocity. When $v_c > v_p$, the weakest, least stable, unit unfolds first, regardless of its position in the chain. Otherwise, the unit at the pulled end unfolds first despite the fact that it may not be the weakest unit alongside the chain. This phenomenon occurs under explainable conditions: at low speeds, the system remains nearly equilibrium, starting the unfolding from the weakest spot, while at higher speeds, force gradients develop forcing the pulled

unit to experience the greatest stress, unfolding first. It is also demonstrated in the article, that for systems with more than one weak spot, multiple critical velocities can arise, leading to different unfolding pathways at different speed intervals.

It is important to note that protein unfolding is influenced not only by the pulling method but also by the type of protein itself. Using Steered Molecular Dynamics (SMD) simulations under constant velocity pulling, researchers have compared alpha-helical and beta-helical proteins to investigate how structural geometry affects energy dissipation during mechanical unfolding. The results [7] showed that alpha-helices tend to unfold more uniformly, with sequential hydrogen bond breakage. In contrast, beta-helices exhibit higher extensibility and energy absorption due to their unique geometrical structure, which allows the backbone and hydrogen bonds to rotate and align with the pulling direction, particularly during later stages of unfolding. These findings highlight the significant role of protein structure in determining unfolding pathways.

Models Developed for Protein Unfolding

As a matter of fact, there are plenty of ML models developed in this field, especially after the release of AlphaFold, an AI system developed by Google DeepMind that predicts a protein's 3D structure from its amino acid sequence. Understanding the static and folded structures gives a hint in revealing the unfolding pathway of proteins. However, it does not capture protein folding and unfolding over time, indicating that this field still is open for development.

One of them is Protein Rapid Estimation of Strength with Deep Learning Models (PRESTO) [13], a novel end-to-end model, which is designed to predict the mechanical strength (F_{max}) of protein sequences. Instead of traditional time-consuming and computationally expensive molecular dynamics methods, this approach requires only amino acid sequences of proteins, making the overall method efficient for large-scale applications in diverse fields. PRESTO is trained, using data from Biomolecule Stretching Database (BSDB) [24], which includes coarse-grained

simulation-based F_{max} values. BSDB contains data on the stretching simulations of 17,134 proteins, derived from PDB. By performing the simulations, researchers have estimated the mechanical resistance of proteins to force, needed for unfolding. Using force-displacement curves obtained from simulations, F_{max} and related unfolding pathway information are provided, including individual mechanical clamps, structural motifs responsible for resisting force. This database incorporates data from other major databases like CATH [19] and SCOP [27], providing information on the evolutionary relationships between proteins and their domains, along with providing biological functional information from Gene Ontology. However, as BSDB contains simulation results, mimicking the real-life stretching experiments conducted via AFM, the results cannot fully reflect the reality and are not fully valid for estimating protein stability.

Although PRESTO offers a powerful tool, leveraging 1D convolutions, bidirectional LSTM layers, and dropout regularization for more accurate predictions, it can be used for large-scale screening of proteins, but is not completely reliable with the necessity to recheck obtained findings.

Another way of predicting protein unfolding is via determining topological graphs, illustrating structural features of proteins, which heavily influence unfolding rates. By examining graph-based representation of proteins, correlations between unfolding rates and several topological measures, including average path length, are measured along with a novel metric, introduced by the authors [9], impact of edge removal per residue. As a result, the impact of contact order, average path length, and clustering coefficient as part of topological measures was negligible compared to the impact from edge removal per residue, which evaluates how sensitive the contact network of a protein is to the loss of connections in unfolding. This means that not only mechanical conditions and protein class play a significant role during unfolding of proteins but the durability of the contact network as well, serving as a breakthrough in understanding and anticipating protein unfolding behavior.

Not only LSTM layers and Topological Graphs are developed in this field but also Convolutional Neural Networks (CNNs) are utilized for the field of molecular biophysics. DeepGOPlus [10] is a DL model, developed for predicting protein functions from amino acid sequences. It is an improved version of previous DeepGO, which had a major limitation in sequence coverage and heavy dependency on interprotein interaction. It was trained on data from SwissProt and CAFA3 (Critical Assessment of Function Annotation) challenge. The model architecture integrates CNN, detecting functional motifs in protein sequences, and sequence-similarity approach, transferring annotation of known proteins to the new sequences. However, regardless of the function prediction and estimation of Fmax, this model does not fully target protein unfolding pathway prediction.

Graph Neural Networks (GNNs) are not spared in the field of biophysics. GSnet [28] is a newly developed and published model designed to predict the physicochemical and geometric properties of proteins based on their three-dimensional (3D) structures. Those properties include salvation free energy, molecular volume, diffusion constants, and hydrodynamic radii. Moreover, GSnet was utilized for predicting solvent-accessible surface area (SASA) and residue-specific pKa values, outperforming EMS-2 and other GNNs. Across all predicted values, salvation free energy presented the least accurate predictions, allowing scientists to improve their model in future. Despite the closeness of the idea to this project’s target, this approach cannot be employed due to its independent nature.

As any prediction can be estimated, analytical models present to be a powerful tool, lacking some accuracy and real-life unpredictability. Muñoz [16] and Eaton introduced a statistical mechanical model that predicts protein folding kinetics using 3D structures of proteins. By deriving essential components from those structures, simplifying the challenge of protein folding and unfolding by assuming the folding to start in local regions, fusing together to turn into native structures. Complex approximations, including single, double, and triple native stretches allowed per molecule, are used to handle the vast conformation space,

expressing the results in one-dimensional vectors of the number of native peptide bonds. They further served as free energy profiles, resulting in prediction of two minima values for proteins with one-step unfolding and more in case of existing intermediate foldings. This approach is computationally complex and includes defined formulas for calculations, while in reality more factors affect the folding of proteins.

Models Chosen for the Project

In the process of thorough search, three main models have been selected to be implemented in the scope of this project. In this section, all three models are described in a more detailed way.

The first model is the Evolutionary Scale Model-2 (ESM-2) [11], a powerful transformer-based protein language model family, which is trained up to 15 billion parameters and millions of protein sequences, using masked language modeling objectives. Its performance was improved compared to its previous version in terms of Template Modeling (TM) Score, higher accuracy with lower perplexity and general efficiency in terms of time and computation.

The second one is ProteinBERT [3], a deep learning language model, designed primarily for protein sequences and developed as a branch of BERT models. Learns both sequence- and protein-level structural and functional annotations and enables transfer learning for diverse custom tasks. By incorporating local and global representations of proteins, ProteinBERT achieved comparable results with larger models, based on different evaluation criteria.

The last model is SciBERT [1], a pretrained language model, developed on the basis of BERT, the same way as ProteinBERT. Its purpose is to address the lack of labeled scientific data, being trained on vast amounts of scientific publications, by improving the scientific Natural Language Processing (NLP) tasks. The model outperformed BERT and currently serves as a strong foundation for processing scientific texts.

Methodology

Python programming language was chosen for analysis, and primarily **Google Collab** services were utilized as a working space for this project. The entire project consists of several notebooks, divided into processing of the dataset and modeling.

Dataset selection and Data Collection

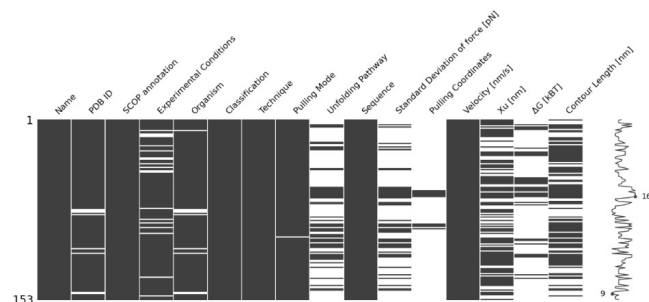


Figure 2: Partial data distribution of MechanoProDB data with identified protein sequences, indicating the amount of missing values in columns.

MechanoProDB [15] was used as a basis of this project. It is a dataset, containing data about mechanical properties of proteins along with experimental conditions, where experiments have been conducted and protein-specific information obtained from PDB. Each entry was retrieved from papers, filling up the columns of the dataset with the results claimed in corresponding papers. Overall, at the moment of usage, the dataset contained 293 entries and 34 columns, filled in by the authors and users, annotating their individual experimental results. This data was further cleaned and subsetted for processing and usage.

Another source, containing data regarding the thermodynamic properties of proteins and their mutants, is ProThermDB [17]. It contains 32,281 entries and 21 columns, describing the protein information, with corresponding PDB and UniProt IDs, allowing extraction of relevant information from their databases, experimental conditions, literature information, and subsequent experimental thermodynamic data. The dataset was not publicly available, so the authors have been contacted along with their representatives to receive the dataset. As this was not a simple process, it took time, which made it difficult to use the data for most of the project’s implementation,

such as model development.

Preprocessing

Since MechanoProDB included data from papers published at different timepoints, the reported data required thorough validation. During the validation process, numerous inaccuracies were identified regarding the reported parameters. Thus, each article has been carefully read and the relevant properties were checked and corrected when necessary. Some parameters were found in figures without direct mentioning in the text, while others were extracted from the written content of the papers. However, as not all papers aligned with the goals of this project, many articles lacked relevant information and were consequently excluded from the analysis.

This project focuses on predicting protein kinetics for single-molecule force spectroscopy (SMFS) experiments, specifically involving Optical Tweezers and AFM. Hence, any data entries derived from simulation experiments were removed from the analysis. The parameters reported for simulations differed significantly from those obtained from SMFS experiments, potentially leading to more variance and noise in data, ultimately not satisfying the objectives of this research.

Regarding ProThermDB, given the larger amount of available data, rows lacking values for the numeric columns relevant to this study were excluded from the analysis. Since this dataset contains experimentally derived data obtained through methods such as differential scanning calorimetry (DSC), circular dichroism (CD), fluorescence spectroscopy, and other mass spectrometry (MS)-based high-throughput techniques, no additional data entries were removed based on the experimental method.

Because the prediction targets differed between ProThermDB and MechanoProDB, all rows missing information for the target variable were also removed from the analysis. Furthermore, it was observed that some entries contained inconsistencies in ID assignment, implying that proteins were saved under another protein’s ID. Several of these entries were manually corrected. Nevertheless, due to the extensive number of data points, it was not

feasible to validate every entry individually, which may potentially introduce sources of error, affecting the model prediction results.

Data Analysis

A preliminary Exploratory Data Analysis (EDA) was conducted to assess the quality of the datasets and examine the general distributions of their columns. The results indicated that the feature importance scores for the target variables, calculated using a Random Forest model, were relatively low, not exceeding a threshold of 0.5. Consequently, features with a non-zero score were retained for further analysis. Principal Component Analysis (PCA) was applied, combined with clustering techniques such as K-Means and DBSCAN, to investigate the potential separation of points. However, most data points aggregated into a single cluster even after the removal of outliers. Attempts to find the optimal number of clusters using the Elbow method did not yield meaningful results, preventing the exclusion of any rows or columns based on clustering.

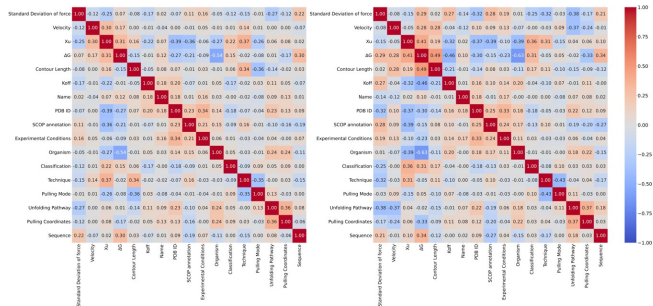


Figure 3: A. Pearson Correlation Heatmap for MechanoProDB; B. Spearman Correlation Heatmap for MechanoProDB.

The distributions of the numeric and categorical columns were examined to get a better insight about their range and frequency, respectively. Due to the limited informativeness revealed by this analysis, several columns were completely removed from the analysis. Furthermore, simple correlation analysis was performed across both categorical and numeric variables. Pearson and Spearman correlation coefficients (Figure 3) were calculated to account for different types of data distributions and relationships. As a result, low correlation coefficients between features were

observed, displayed in a form of correlation heatmaps. These correlations were checked for columns, further included into the analysis.

Sequence Extraction

Since the primary objective of this project was to predict energy kinetics of proteins based on their sequences, corresponding protein sequences were derived for each row by utilizing a combination of columns such as 'Name', 'PDB ID', 'PDB_wild', 'UniProt_ID', 'Total Length', 'Experimental Conditions', 'Domain Coords', 'Authors', 'Pulling Reference' and other auxiliary columns if present.

The initial step involved developing an automated script that, given a PDB ID, would query the database through an API and retrieve the corresponding sequence from FASTA files, widely used in bioinformatics and biochemistry, removing the headers in the process. However, several challenges emerged, making this approach ineffective for MechanoProDB. Some proteins had multiple FASTA files associated with them, often with unpredictable naming conventions, while others lacked PDB or UniProt ID entries, preventing the automation of this process. In certain cases, the structures associated with PDB or UniProt IDs were incomplete or modified due to mutations introduced into experimental designs, complicating the sequence extraction.

Consequently, it was decided to extract the sequences for MechanoProDB manually, despite the time-consuming nature of the process as it was the only possible option. For each row, the PDB ID was used to find the entry in the PDB database if present. The retrieved structure and sequence were compared to the reported 'Total Length' (in amino acids). If the sequence length matched or was approximately equal (within ± 5 amino acids), further verification was performed by analyzing the 'Name' and 'Experimental Conditions' columns to detect any reported mutations in the sequence. If mutations were identified, the sequence was modified accordingly to accurately reflect the protein construct used in the original experiment for obtained parameters. In cases where the lengths differed significantly, the 'Domain Coords' column, containing information about the range of

amino acids used in the experiment, was employed to extract the correct subsequence.

However, a considerable number of sequences either did not correspond to the reported amino acid lengths and did not contain 'Domain Coords' value or lacked complete information for both 'Total Length' and 'Domain Coords' columns. In such cases, the 'Authors' and 'Pulling Reference' columns were used to find the original publications, describing the conducted experiment. Those articles were then examined for additional information regarding the experimental sequence, such as domain coordinates, the sequence itself or the alternative PDB or UniProt IDs. If no reliable sequence could be obtained after this review, the corresponding row was excluded from further analysis.

Additional validation steps were used to ensure the correctness of the sequences. Derived sequences were input into the AlphaFold web application to predict their structures, which were subsequently compared to the 'Unfolding Pathway' value describing the number of intermediate states during unfolding. If the protein had only one unfolding step but the structure showed multiple unfolding states, this sequence was rechecked and if not corrected, dropped from the analysis. Some experiments involved either the combination of multiple proteins or repeated use of the same protein, adding further complexity to the verification process.

As for ProThermDB, the initial strategy of automated sequence extraction via database queries proved to be highly effective. Sequences were retrieved from both the PDB and UniProt databases based on the availability of corresponding IDs and were recorded into a working copy of the original dataset. Nonetheless, during this process, several proteins' extraction failed due to the deletion or deprecation of their entries in the database. For such cases, either the sequences were manually extracted from the PDB if a substitutive ID was present in the database or were excluded from further analysis.

Feature Selection

Following the thorough analysis of the MechanoProDB and ProThermDB, several

features were excluded from further analysis due to their limited range of values, lack of direct relevance to the prediction targets, or insignificance for model performance.

For MechanoProDB, the initial columns removed included 'Authors', 'Pulling Reference', 'Released' as they contained metadata related to publication, which had already been utilized for sequence extraction and parameters validation. These columns held no additional predictive importance. Further columns were excluded due to their sparsity, redundancy, or irrelevance to the project's objectives, particularly if those columns were already utilized for retrieving other data for further usage, such as sequence and parameters extraction.

After conducting the Feature Importance analysis using Random Forest, additional columns were discarded, including 'F05 [pN]', 'Mechanical Clamp', 'Unfolding Annotation', 'DOI Structure', '3D Structure', 'Mutations', and 'Domain Coords'. The 'Temperature' was also removed, as it contained only two values (20 and 23 °C) and had no direct influence on the predictions. Similarly, although 'Loading Rate' contained 84 unique values, it was not directly associated with the unfolding processes in the scope of this project and was therefore excluded. Furthermore, the 'PDB ID' and 'Uniprot' columns were merged into a single column called 'PDB_UniProt', as those two columns were mutually exclusive.

In the case of ProThermDB, a large number of columns were discarded due to redundancy or their irrelevance in the analysis. The key columns, which were left, contain information regarding the types of proteins, the organism of origin, secondary structure annotations, and the melting temperature for proteins, which served as the primary prediction target. Additionally, the 'pH' column was retained, as it provided meaningful information about the experimental conditions and, unlike other excluded columns, did not represent relative changes in parameters between the wild type of proteins and their mutated versions.

Feature Engineering

MechanoProDB contained a significant amount of missing data across both numerical and categorical

features. Thus, handling two types of data was required.

Initially, several columns were merged or newly created to better organize the information. The columns 'Koff' and 'Unfolding Rate at Clamp Force' were merged into 'Koff' due to their similar concept: one is for unfolding rate at zero force, while the other one captures it at a specified force. This newly created column was included among the other target variables to predict. Missing values in the 'Total Length' column were filled using the lengths of the obtained sequences, which subsequently allowed the calculation of 'Contour Length' (the fully unfolded protein length in nanometers) if the value was missing. Based on the average observed relation between contour length and sequence length, each amino acid was estimated by 0.35 nm.

Afterwards, a 'Pulling Coordinates' column was introduced due to its importance in the unfolding process, as the pulling points, specifically amino acids, significantly influence both energetic parameters and unfolding mechanisms. When available, specific start and end points were extracted from the original publications. Otherwise, by default, the pulling was assumed to occur from N- and C-termini (starting and ending amino acids) of the sequence. This information was then divided into two separate columns, 'Pulling Start' and 'Pulling End', for the ease of usage in the models.

For the next step, imputing missing categorical data was initiated. The 'Unfolding Pathway' column, which indicates the protein's unfolding mechanism and intermediate states, was reconstructed if possible based on the original structure reported in the dataset. Entries as N_U[18.5] and N_I[15]_U[16] were interpreted accordingly, with 'N' representing the native folded state of the protein, 'I' for Intermediate state of the protein, and 'U' for the fully unfolded state of the protein. The numbers in parentheses indicate the contour lengths at different unfolding stages. Thus, to fill this column, no estimation could be made for the intermediate states; however, if the protein was small, it was assumed to unfold in a single step. A threshold of 110 was chosen, meaning that any protein with a 'Total Length' less than 110 would be assigned the

value N_U['Contour Length'], while others would be assigned the <UNK> token, which is widely recognized by models as representing an unspecified value and is handled accordingly.

Using the final 'Unfolding Pathway' values, domain subsequences were estimated by partitioning the protein sequence according to unfolding transitions. Ideally, separate parameters should have been documented for each subsequence of the original sequence, as containing unique amino acids leads to different predicted results. However, due to the lack of documented separate parameters in the original papers, the full sequence was treated as a single unit for prediction.

Following the imputation of categorical features, tokenization was performed using three different models: ProtBERT [3] for protein sequences, SciBERT [1] for scientific text columns, and experimentally ESM-2 [11] for full-text columns requiring its specific tokenization. Although tokenizing text columns by ESM-2 tokenizer designed for protein sequence was not the optimal approach, it was implemented experimentally, remarkably displaying the best results.

For numeric columns, regression-based imputation [26] was employed. Initial attempts using median and mean imputations yielded poor model performance, though being widely used for missing values handling, likely due to inaccurate assumptions these simple methods impose on experimental data. Regression imputation leveraged relationships between variables to predict missing values, proving to be more appropriate and effective in this particular case.

As a result, the cleaned numeric features were gathered into a new 'numeric_embeddings' while the cleaned scientific text columns were joined into a single 'Joint_Text_Cols' column, subsequently tokenized using SciBERT. The target variables - ΔG (free energy difference between the folded and unfolded states), X_u (distance along the reaction coordinate between folded and transition states under force), and K_{off} (rate constant of unfolding) - were combined into a single 'targets' column for model training.

In contrast, for ProThermDB, the dataset's larger size allowed the selection of a subset containing

only complete numerical data, avoiding the need for numeric imputations. Categorical entries containing missing values were replaced with the <UNK> token. The same tokenization and feature aggregation strategy applied to MechanoProDB was also used for ProThermDB. Specifically, the ‘targets’ column was created to include the melting temperature (T_m), and numeric features were compiled into the ‘numeric_embeddings’ column in the same manner to keep the structure consistent.

Models Pipeline

Four distinct models were developed for the task of predicting protein sequences. All models shared a common strategy: combining the tokenized protein sequence with scientific text columns and numeric features as inputs for the models pipeline. Three of the models employed the same tokenization method across all textual features, while the fourth model utilized its own specialized tokenizer specifically designed to process the text features, mostly protein sequences.

During training, a Learning Rate Scheduler was implemented to dynamically adjust the learning rate based on the validation loss for more stable convergence. Additionally, Early Stopping was applied, stopping training if the validation loss did not improve for 30 consecutive epochs to prevent overfitting. The best model estimated based on the difference between the validation loss is saved after each epoch of training to keep the parameters of the best model for future inference.

The Adam and AdamW optimizers were utilized to update the model parameters, based on the best performance for that model, chosen for their ability to efficiently handle sparse gradients and adapt learning rates during training. For each model and dataset, training metrics - including training loss, validation loss, and evaluation metrics - were recorded at every epoch and converted into a structured dataframe. Finally, all trained model artifacts, including weights, scalars, and tokenizers, were saved for the subsequent inference stage.

ESM-2 with Regression Head

For this model, the ESM-2 architecture with 8 million parameters was initially selected, primarily

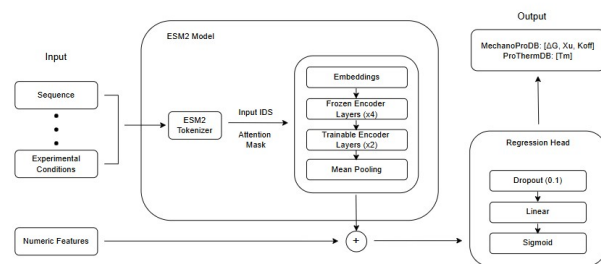


Figure 4: ESM-2 Model Architecture, fine-tuned for multi-output regression task.

due to the relatively small size of the main dataset used for this project’s model development (MechanoProDB). Both the pretrained base model and its corresponding tokenizer were initialized to be incorporated into the models pipeline.

To leverage the advantages of transfer learning, the majority of the ESM-2 model was frozen, preserving the learned representations from its training on large protein data. However, the last two encoder layers, out of total six, were unfrozen and made trainable for our particular datasets. This strategic partial unfreezing aimed to allow fine-tuning of the model on the specific regression task and learning particular distribution of our data from datasets, along with preventing the risk of overfitting due to one of the datasets’ limited size. It enabled the model to adapt higher-level protein representations to the current purpose while maintaining the strength of previously learned knowledge.

All text columns - including the protein sequence and additional scientific description columns - were tokenized using the specific tokenizer associated with ESM-2. These tokenized outputs were subsequently passed through the ESM-2 model to generate high-dimensional embeddings. Afterwards, Mean Pooling was applied to the token-level embeddings output to get a fixed-size representation by averaging across the token dimensions.

Separately, numeric features were extracted, standardized using StandardScaler, and prepared for integration. The numeric features were then concatenated with the embeddings output by ESM-2, forming a unified feature vector combining textual and numeric information. This combined representation was passed through a Regression

Head, consisting of a Dropout layer (with a dropout probability of 0.1) to prevent overfitting, followed by a fully connected (linear) layer to project the features into the target dimension of three. Finally, a Sigmoid activation function was applied to bound the model outputs between 0 and 1.

Because the targets were scaled using MinMaxScaler to a $[0, 1]$ range, the use of the Sigmoid activation function ensured alignment between the output range of the model and the scaled targets, improving the training stability and interpretability during evaluation.

Combined ProtBERT and SciBERT models

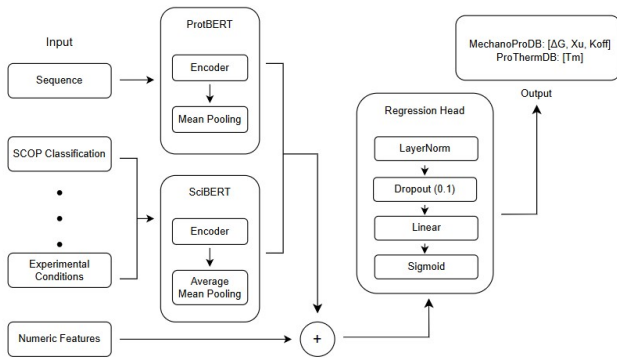


Figure 5: Combined ProtBERT and SciBERT Model Architecture with separate regression Head.

This particular model integrated the combined outputs from ProtBERT and SciBERT tokenizers. The protein sequence was tokenized using ProtBERT, while the scientific text columns were tokenized with SciBERT, each tailored to their specific data types. Mean pooling was applied independently to the tokenized embeddings from both models to extract fixed-size feature vectors. The resulting pooled embeddings were then concatenated with the numeric feature vector to form a unified input representation. This combined vector was passed through the same Regression Head, as for ESM-2 (Dropout, Linear, Sigmoid). Since the targets' scaling is the same for three Neural Networks, the same approach was used for this specific model.

BiLSTM model

For this model, the same tokenization strategy was employed as in the Combined ProtBERT and SciBERT model: the protein sequence was

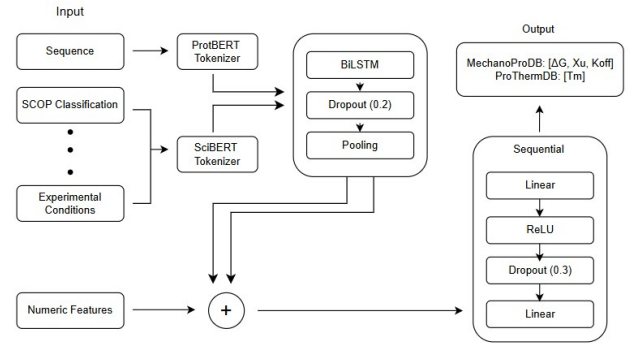


Figure 6: BiLSTM Model Architecture, combining ProtBERT and SciBERT tokenizations.

tokenized separately using ProtBERT, while the remaining scientific text columns were tokenized using SciBERT. This separation allows each domain-specific tokenizer to effectively capture the linguistic characteristics of its respective input.

The resulting embeddings from each encoder were passed into separate Bidirectional LSTM (BiLSTM) layers, which help take sequential dependencies in both forward and backward directions. To prevent overfitting, a dropout layer with a probability of 0.3 was applied to the BiLSTM outputs.

Instead of using conventional pooling methods, the model applied last hidden state pooling by concatenating the two last hidden states (forward and backward) from the last LSTM layer. This technique was chosen due to its ability to retain the most significant information from both directions.

These pooled embeddings were then combined with the numeric features, forming a comprehensive feature vector. This vector was passed through a fully convected sequential head consisting of four layers: a linear transformation layer, a ReLU activation function, a dropout layer (with a rate of 0.3), and a final linear layer that outputs the predictions. Unlike previous models, a sigmoid activation function was not applied at the output stage. This was an experimental approach to check the predictions precision. The target structure and scaling strategy remained consistent with those used in the prior models.

TensorFlow Decision Forests

The final model employed for this task was based on TensorFlow Decision Forests (TFDF). TFDF enables the use of tree-based models - such as Random Forests and Gradient Boosted Trees - directly within the TensorFlow ecosystem. Decision Forests were selected for their interpretability, strength for small datasets (as MechanoProDB), and because they often require less extensive feature engineering compared to neural networks.

For feature preparation, the text columns were tokenized following the same approach as in the previous two models: the protein sequence was tokenized using ProtBERT, while the scientific text columns were tokenized using SciBERT. After tokenization, embeddings were obtained for each input, being averaged to create a fixed-size representation. The resulting embeddings were then concatenated with the numeric features to form complete feature vectors, which were used as inputs to the TFDF model.

To optimize the performance of the decision forest, Optuna was integrated for hyperparameter tuning, searching over parameters such as minimum number of examples per node and maximum tree depth.

Metrics used

- **MSE** (Mean Squared Error) measures the average of the squares of errors and is calculated by the formula below:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- **RMSE** (Root Mean Squared Error) measures the root mean square deviation between the original and predicted values and is calculated by the formula below:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}$$

- **RRMSE** (Relative Root Mean Squared Error) measures RMSE relative to the scale of data, making it more interpretable, and calculated by the formula below:

$$\text{RRMSE} = \sqrt{\frac{\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^n (\hat{Y}_i)^2}}$$

- R^2 represents the proportion of the variance for a dependent variable that is explained by an independent variable and is calculated by the formula below:

$$R^2 = 1 - \frac{\sum_i (Y_i - \hat{Y}_i)^2}{\sum_i (Y_i - \bar{Y})^2}$$

- **Validation Loss** (scaled) measures how good model performs on data not seen before (validation set) based on MSE metric.

Structure

The entire structure of this project is described in a separate ReadMe file, which can be found in the zipped folder for this project. The general structure of the components notebooks contain is described along with each notebook's purpose. Composition of folders and files is presented along with the structural outline. Refer to that file for usage and running the codes.

Results

MechanoProDB

For MechanoProDB, the relatively small dataset size and, thus, faster training time, allowed more training epochs compared to ProThermDB. Specifically, 50 epochs were run on 101 data entries, with the remaining 26 entries used for validation, following a 80/20 train-validation split.

To assess models performance, a series of plots for validation data was created, using the metrics DataFrame for each model saved after their separate training. Those DataFrames have identical structure, making the code consistent with reading them. These validation plots visualize the same performance metrics across all models - three neural networks - enabling a direct comparison between the models to observe their predictive capacity across all target variables. Each model was assigned a distinct color for clarity: red for fine-tuned ESM-2, blue for the combined ProtBERT

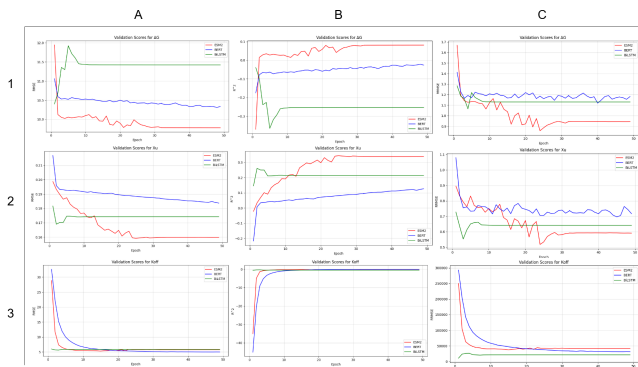


Figure 7: Visualizations of evaluation metrics (A. RMSE, B. R^2 , C. RRMSE) for MechanoProDB target variables (1. ΔG , 2. X_u , 3. K_{off}) prediction, using three different models (colors: red - ESM-2, blue - BERT (by short), green - BiLSTM).

and SciBERT model (shortly labeled ‘BERT’), and green for the MultiInputBiLSTM model (shortly labeled ‘BiLSTM’). TFDF was plotted separately via using the logs and RMSE as a basis for evaluation.

1. ΔG

Focusing on the ΔG target variable, ESM-2 clearly outperforms the other two models, demonstrating the most stable and effective learning behavior. It converges more rapidly compared to both BERT and BiLSTM, consistently achieving the lowest error values and the highest R^2 score, indicating its ability to capture meaningful variance in data. Its RMSE stabilizes under 10, particularly around 9.5, though the decreasing trend suggests that model could converge to better results with additional training epochs. In contrast, the BiLSTM model performs poorly, plateauing pretty early at a higher error level and showing insignificant improvement, which reflects limited generalization capacity and underfitting. BERT ranks second but still fails to generalize effectively, with moderate errors.

2. X_u

For this target variable, ESM-2 again outperforms the other two models, achieving the lowest RRMSE not only for this specific prediction but across all targets and models. It steadily reduces its error with the increase of epochs. However, its best performance is observed around the 25th epoch rather than at the last epoch, indicating faster convergence but also suggesting potential

overfitting and space for improvement through gathering more data as an input. While BiLSTM initially shows promising performance with rapid convergence, it plateaus very fast within less than 10 epochs and fails to improve its performance further, reflecting weaker generalization capacity. In contrast, ‘BERT’ demonstrates an increasing tendency for R^2 and decreasing one for RMSE, reaching its best performance at the final epoch, suggesting that additional training epochs could improve its performance and possibly reach a better state than ESM-2.

3. K_{off}

As the final target analyzed, Koff proved to be the most challenging variable to predict, with all three models performing poorly. Potential causes of that could be high variance in data and limited number of data entries, leading to noisier predictions and worse performance. Despite the challenges, the BiLSTM model slightly outperformed the others in terms of RRMSE, although the values remained very high, indicating that predicting the mean for K_{off} would be more efficient than running neural networks. The initial instability and persistent error show that, without more data and improved models, this target will likely remain the most difficult one to predict.

In conclusion, ESM-2 consistently outperformed other two modes in 2 out of 3 target variables, suggesting future refinement of this particular model. Along with the previously described metrics, ESM-2 also has the lowest validation loss among all models, indicating stronger generalization capacity and learning ability. Its ability to capture sequence-level representations is stronger, indicating the need to potentially combine ESM-2 sequence tokenization with SciBERT or other scientific text tokenizers to achieve better results.

For the TFDF model, the results cannot be visualized across epochs as with the neural networks, and were instead introduced in a form of table. The evaluation shows that TFDF achieves relatively higher R^2 scores for ΔG and K_{off} , which are harder to predict, but is less for X_u compared to ESM-2. In terms of RRMSE, neural networks outperform TFDF across all target variables, while

Metrics	MSE	RMSE	RRMSE	R ²
ΔG	90.26	9.50	0.89	0.13
X_u	0.03	0.16	0.57	0.33
Koff	19.71	4.44	17695.18	0.15

Figure 8: Table with validation metrics for three target variables, obtained by TFDF model.

other evaluation metrics are either slightly better for neural networks or are nearly the same. This suggests that neural networks will perform better with more data entries and are likely to surpass TFDF in overall predictive performance across all evaluation metrics in future.

ProThermDB

Following the training of the developed models on MechanoProDB, specifically adjusted for smaller datasets, the same models were applied to a larger dataset to evaluate their ability for generalization. Due to technical difficulties during the runs on AUA machines, specifically frequent power fluctuations leading to results loss within a week), only two out of the three neural networks could be executed on PoThermDB. Additionally, the task was not well-suited for TFDF, limiting its applicability. However, this paper reports the results for only one model, which is ESM-2, which had previously demonstrated the best performance on MechanoProDB. While optimal performance was not expected by applying models fine-tuned on smaller datasets to a larger one, the main objective was to evaluate whether the models could still produce satisfactory results for at least some target variables. With only one target to predict, which is the melting temperature of the proteins, the models were easier to evaluate.

The findings indicate that, in general, ESM-2 maintains relatively stable performance on the larger data even without modifying any of the fine-tuned hyperparameters such as optimizer, scaling method, or pooling strategy. RRMSE shows a consistent decreasing trend, dropping from approximately 0.22 to 0.18 at epoch 15, indicating

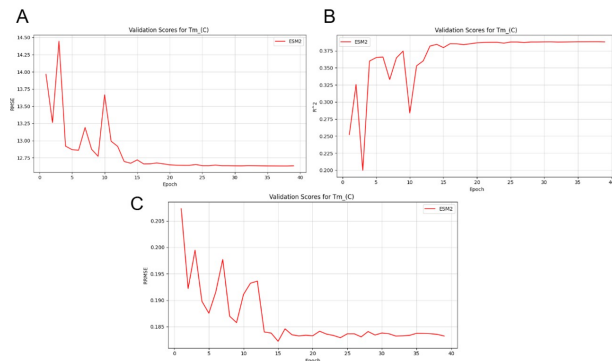


Figure 9: Visualizations of evaluation metrics (A. RMSE, B. R^2 , C. RRMSE) for ProThermDB target (T_m) predictions obtained from ESM-2 model.

training stability. Meanwhile, the R^2 score increases, starting from 0.2 as a minimum threshold and reaching nearly 0.4, indicating that the model is able to explain up to 40% of the variance in validation data. While early fluctuations are observed in R^2 scores, they are typical when training is done on complex datasets, which in this case integrates both tokenized text data and numeric features.

It is important to note that R^2 score, being not the highest, still suggests useful predictive ability, especially considering the unpredictable and noisy nature of unfolding of proteins. As for the last analyzed parameter, RMSE starts at a relatively high value, showing a clear downward trend, reinforcing the model’s effective learning and convergence.

In conclusion, despite the fact that the models were originally fine-tuned and developed for a smaller dataset, the ESM-2 model still shows relatively satisfactory results on a larger dataset without key adjustments. This suggests efficient performance and a solid ground for future model development, leading to greater accuracy with more specific optimization.

Inference

Since the ESM-2 model outperformed the other three models, the inference pipeline was developed specifically for this model to predict three target variables: ΔG , X_u , and K_{off} , based on training on MechanoProDB. The saved state_dict of the best ESM-2 model is loaded along with the tokenizer used for text features and scalers used for numeric

features during training.

All input columns are user-defined, meaning that the user needs to provide any known feature values to get the predictions, while inputting the protein sequence is a must. If a text input is missing, it is automatically replaced by '<UNK>' token, recognized by the model. For numeric features, if missing, the mean values from MechanoProDB are used as defaults for their corresponding columns. This approach ensures that no undefined values are passed to the model for making predictions for a single protein.

Plots

Once the three target values are predicted, they can optionally be used as a basis to plot the Free Energy Profile and Force-Extension curve, describing that protein's unfolding. These visualization steps are not mandatory: if the user wants to generate the plots to get the idea of unfolding behavior, plotting functions can be called, which automatically use the predicted values. Alternatively, the user may manually input custom values instead of predicted variables to plot the images based on arbitrary data.

The first plot (Figure 1 C.) illustrates the Free Energy Landscape (Profile) of a protein undergoing mechanical unfolding. It is a classic one-dimensional projection of the protein's free energy surface along the reaction (pulling) coordinate. This plot gets as an input only two of the three predicted values, which are ΔG and X_u , as the chosen formula from [6] does not incorporate it into the model.

$$G_0(x) = \frac{\Delta G^\ddagger x}{\mu x^\ddagger} \left[1 - (1 - \mu) \left(\frac{2x^2/x^\ddagger}{|x| + x \sin \theta} \right)^{\mu/(1-\mu)} \right],$$

where $\Delta G^\ddagger$ and $x^\ddagger$ are the predicted ΔG and X_u , μ determines the kinetic brittleness and $\mu \in (0, 1)$, while θ indicates well-barrier asymmetry and $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. $k_B T$ is by default set to 4.1 pN·nm, assuming room temperature for all experiments. Here, k_B is the Boltzmann constant, equal to 1.380649×10^{-23} J/K.

The default parameters, which are $\theta = 0$ and $\mu = \frac{2}{3}$ are used, according to the conventional standards in free energy surface modeling. However, those values can be adjusted by the user if custom plots

are needed. The points G_n and G_t from the plot represent the free energy in the native and transition states respectively, visually showing the barrier protein must overcome to unfold under applied force.

The second plot (Figure 1 B.) shows a Force-Extension, which represents how proteins respond to mechanical stretching. These plots are generally obtained using SMFS techniques, such as AFM and Optical Tweezers, while in this particular case they are obtained from model predictions using analytical approximations from [20] and [21].

$$x(f) = \frac{4}{3} - \frac{4}{3\sqrt{f+1}} - \frac{10e^{\sqrt[4]{900/f}}}{\sqrt{f} \left(e^{\sqrt[4]{900/f}} - 1 \right)^2} + \frac{f^{1.62}}{3.55 + 3.8f^{2.2}}$$

$$f = \frac{Fp}{k_B T}$$

$$F \approx \frac{\Delta G^\ddagger}{\nu x^\ddagger} \left(1 - \left| 1 - \frac{k_B T}{\Delta G^\ddagger} e^{\frac{k_0 k_B T}{x^\ddagger \dot{F}}} E_1 \left(\frac{k_0 k_B T}{x^\ddagger \dot{F}} \right) \right|^\nu \right) \cdot \theta (\dot{F}_{\text{cFirst}} - \dot{F}) + \left(\frac{\Delta G^\ddagger}{\nu x^\ddagger} - \sqrt{2\dot{F}_{\text{cFirst}} x^\ddagger \zeta} + \sqrt{2\dot{F} x^\ddagger \zeta} \right) \cdot \theta (\dot{F} - \dot{F}_{\text{cFirst}})$$

$$\dot{F}_{\text{cFirst}} = \frac{k_0 k_B T e^{\frac{\Delta G^\ddagger}{k_B T} + \gamma}}{x^\ddagger} \left(1 - e^{-\frac{\Delta G^\ddagger}{k_B T}} \right)$$

$$k_0 = \frac{3\Delta G^\ddagger}{\pi (x^\ddagger)^2} e^{-\frac{\Delta G^\ddagger}{k_B T}}$$

$$\theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

where k_0 , $\Delta G^\ddagger$, $x^\ddagger$ are the predicted target variables k_{off} , ΔG , and X_u , ν is the free energy profile defining factor (with $\nu = \frac{2}{3}$), $\gamma \approx 0.5772$ is the Euler–Mascheroni constant, ζ denotes the diffusion coefficient, and P is the persistence length, conventionally set to 0.4 nm. The last parameter is $\dot{F}$, indicating the loading rate, which is by default set to 10^5 pN/s.

This plot visualizes how much force is needed to unfold the proteins and how their domains influence mechanical stability. Multiple sequential drops are observed in complex proteins and indicate several protein unfolding domains.

The plot is produced through a vast series of computations, involving eight interconnected functions. For the final call of the function, no parameters are required for specification until the user wants to place their own custom values to simulate different unfolding conditions and observe the changing behavior.

Discussion

Throughout this project, multiple challenges were faced, being either partially resolved or temporarily neglected due to the project's scope. These ranged from novelty of the idea to various technical obstacles met during the realization of the project.

1. **Novelty of the Idea:** one of the central challenges was the innovative nature of the research. As already highlighted earlier, predicting the mechanostability of proteins is a relatively unexplored area. While related topics have been studied by scientists, no prior work has directly addressed this task, as confirmed by an extensive literature review. Existing approaches rely heavily on conventional protein assessment methods, leaving this field open for advancement and future development. However, it has its own drawbacks: the absence of pilot models or guidance from the scientific community meant that extensive experimentation was required, consuming significant time and effort and limiting the optimization of final results.
2. **Small Size of the Initial Dataset (MechanoProDB):** MechanoProDB is currently the only publicly available dataset containing experimental measurements of protein mechanostability and unfolding kinetics parameters. Despite its potential, it includes only 293 data entries, further cleaned into only 127 usable entries, making it relatively small for training machine learning models. Extensive efforts were made to identify complementary datasets, which could be combined with MechanoProDB and integrated into the analysis, but no suitable alternatives, published or in development,

were found. Consequently, all modeling approaches had to be optimized for smaller datasets, which may reduce generalizability to larger datasets like ProThermDB, which contains approximately 32,000 data entries.

3. **Data Quality Issue:** Although MechanoProDB initially looked promising, manual verification revealed numerous inconsistencies in reported values. As a result, each entry had to be rechecked against the original publications, over 100 in total, to confirm parameters accuracy and capture experimentally specific details. This process was time-intensive but necessary due to the dataset's small size to ensure data reliability for the models learning. For ProThermDB, due to larger dataset size, it was not feasible to validate the data within a short time period despite the clear signs of errors noticeable at first glance.
4. **Incomplete Data Access (ProThermDB):** After several attempts to contact the authors of ProThermDB, only a partial version of the original dataset was received. Unfortunately, target parameters, such as ΔG were missing from the data. Despite additional efforts, full access is not granted yet, limiting the analysis to available variables like melting temperature of proteins (T_m). However, further attempts will be made to obtain the full dataset, as this project remains ongoing.
5. **Training Time and Scalability (ProThermDB):** Since the models were initially designed and optimized for MechanoProDB, no regulations were made for training on significantly larger datasets. When applied to ProThermDB, training time per model increased from 6 hours of total training on 50 epochs (MechanoProDB) to more than 7 hours per epoch for ProThermDB. Due to these time constraints and the late receipt of the second dataset, the BiLSTM model could not be fully trained and evaluated.
6. **Limited Computational Resources:** Early model development was constrained by the use of a 12 GB RAM environment, where frequent

Google Collab disconnections disrupted long training sessions. The transition to Google Collab Pro+ resolved some issues but eventually ran into resource usage limits. Models were then moved to a local 32 GB RAM machine, which allowed training but required significant time. Additionally, frequent power outages during training led to data loss and further delays in model convergence. This happened due to the multiple electricity interruptions at American University of Armenia and total shutdown of the machines, leading to the results entirely lost without even reaching several epochs of training to be evaluated and presented.

Conclusion and Future Work

This project studied the protein unfolding, with a particular focus on the kinetics of this process, the conventional experimental methods and existing models related to predictions in this field. To address the high cost, in terms of both time and money, of traditional experimental techniques, novel models were developed and fine-tuned to automate the process of obtaining unfolding-related properties.

Among the four trained models, ESM-2 demonstrated the best performance based on evaluation metrics, suggesting that with further optimization, it could deliver better scores with higher precision. This improved model will enable scientists to get relatively accurate and speedy results, applicable across various fields, starting from drug design and finishing with curing diseases, such as Alzheimer disease, appearing as a result of protein misfolding.

Despite the promising results, further improvements are needed, as described in the ‘Results’ section of this paper. Future improvements include experimenting with additional modeling approaches, correctly handling text tokenization and improving current tokenization strategies by integrating various tokenizers and evaluating the scope of changes, and experimenting with alternative optimizers. However, it should be noted that model performance does not only depend on the techniques, but also on the availability of high-quality

data obtained from conducted experiments. Simulations and approximations alone are not sufficient and can potentially lead to unreliable outcomes.

As this is an ongoing project, the following improvements in the flow are planned:

1. Integration of Structural Data

Incorporating protein structural information is a crucial step for more accurate predictions. Protein conformation significantly influences the unfolding behavior, leading to different results for the same sequence. It could explain some current fluctuations in the results.

2. Unfolding Pathway Prediction

Initial steps have been made to integrate Unfolding Pathway prediction, particularly the estimation of the contour lengths of individual unfolding domains in complex proteins. This would enhance the modeling of large, multi-domain proteins, for which prediction of a single set of parameters is incorrect due to different nature of domains.

3. Single Modeling Pipeline

Currently, models trained on MechanoProDB, a small dataset with three target variables to predict, are not directly compatible with ProThermDB, a larger dataset with a single target variable to predict. This means that separate models for each dataset should be developed and subsequently merged into a unified framework.

4. Improved Inference Workflow

Automated parameter extraction from protein sequences or corresponding IDs (PDB/UniProt) will be integrated. By querying external databases (as is currently done for ProThermDB sequence extraction), structural and textual information will be retrieved (e.g., AlphaFold), eliminating the need for extensive manual input and additional effort from the user.

5. Web Application

A user-friendly web application will be developed. Users will need to input only the protein sequence and suggested experimental conditions to get the results. The application will then fetch the associated structural and sequence information, run the models and return the predicted properties

with corresponding visualizations and detailed unfolding metrics, which are described in the Results section.

Appendix

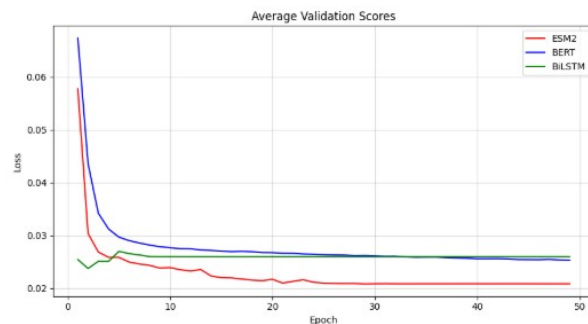


Figure 10: Visualization of Scaled Validation Loss for MechanoProDB predictions obtained from three neural networks.

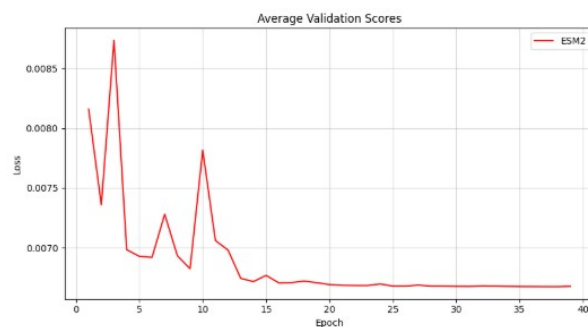


Figure 11: Visualization of Scaled Validation Loss for ProThermDB predictions from ESM-2 Model.

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